101: Java

Apr, 2023

Objective

- ~~インストール: Intellij IDEA, JDK~~
- Hello world!
- 文法
- Data types
- Operators (演算子)
- Loops (繰り返し処理)
- Condition (条件分岐)
- Array (配列)
- Exception (例外)
- Files and I/O
- Scanner

~~JDK~~

- ~~Java Development Kit 17~~
  - https://www.oracle.com/java/technologies/downloads/#java17
  - Windows: x64 Installer or x64 MSI Installer
  - Debian/Ubuntu: `sudo apt update && sudo apt install -y openjdk-17-jdk`
- 環境変数?
  - `java -version`
  - `javac -version`

~~Intellij IDEA~~

- ~~Intellij IDEA Community Edition~~
  - ~~https://www.jetbrains.com/idea/download/~~

MJBK防止策: UTF-8

For example, some Shift-JIS characters include a backslash (0x5C "\") in the second byte, which is used as an escape character in many programming languages.

構		わ		な		い	
8d	**5c**	82	ed	82	c8	82	a2

A parser lacking support for Shift JIS will recognize 0x5C 0x82 as an invalid escape sequence, and remove it.[3] Therefore, the phrase cause mojibake.

高			塢		ネ	い	
8d		82	ed	82	c8	82	a2

https://en.wikipedia.org/wiki/Japanese_language_and_computers

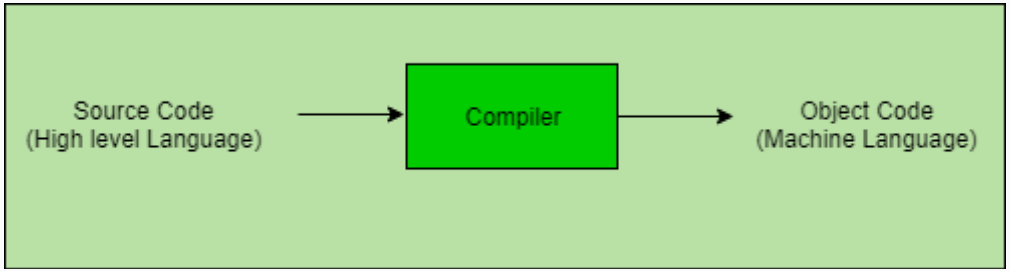
Hello world

```
public class MyFirstJavaProgram {            //class

    /* First java program.
     * This will print 'Hello world' as the output
     */

    public static void main(String[] args) {       //main() method not Main
        System.out.println("Hello world");     /*prints "Hello world"*/
    }
}
```

```
javac MyFirstJavaProgram.java
java MyFirstJavaProgram
```




```
#include <stdio.h>
int main() {
   printf("Hello,
world!\n");
   return 0;
}
```

```
#include <iostream>
int main() {
    std::cout << "Hello world!\n";
    return 0;
}
```

```
print("Hello world")
```

Object-oriented programming (OOP)

- Objects

States(属性/状態)

♂/♀、年齢、白柴/胡麻柴/黒柴, etc.

Behaviors(操作/動作)

- Class
  - blueprint/template (e.g. DNA, RNA)

例: class Dog

```
public class Dog {
  String breed;
  int age;
  String color;

  void bark() {
    System.out.println("Wan wan");
  }

  void eat() {
  }

  void sleep() {
  }
}
```

Keywords: abstract, boolean, break, byte, case, catch, char, class, do, for, if, …

Constructors

MyClass.java

```
public class MyClass {
  int num;
  MyClass() {
    num = 100;
  }
}
```

constructor

ConsDemo.java

```
public class ConsDemo {
  public static void main(String[] args) {
    MyClass t1 = new MyClass();
    MyClass t2 = new MyClass();
    System.out.println(t1.num + " " +
t2.num);
  }
}
```

new objects

Constructors

MyClass2.java

```
class MyClass2 {
  int x;

  MyClass2(int i) {
    x = i;
  }
}
```

instance variable

constructor

parameter

ConsDemo2.java

```
public class ConsDemo2 {
  public static void main(String[] args) {
    MyClass2 t1 = new MyClass2(10);
    MyClass2 t2 = new MyClass2(20);
    System.out.println(t1.x + " " + t2.x);
  }
}
```

- a special method, matches the class name
- does not have a return type (void, int, etc.)
- is called when the object is created
- All Java classes have constructors